CLASS LIST NUMBER	**FACULTY**: INFORMATION & COMMUNICATION TECHNOLOGY **DEPARTMENT**: COMPUTER SCIENCE **SUBJECT**: Principles of Programming and Introduction to Programming (Java) PPB115D/PPG115D/PPBF115D/PPGF15D	**CT5**

Tshwane University of Technology

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Class Test 5A
Duration: 1 hour 00 minutes
Date: 20 October 2023
Total: 33
Pages: 10

Examiner
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Moderator
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Instructions

- When you are done, follow the instructions on how to upload. ***If you do not submit your files on EC, you will NOT be supplied with the opportunity to write a sick test or any supplementary test.***
- Save your work frequently.
- This is a closed book test. **NO** additional resources, like the Internet or EC may be used for help. **You may NOT use any flash sticks during the test.**
- Add your name as a comment to the top and the bottom of your files.
- Please note that **marks will be subtracted** if proper coding conventions are not utilised. (e.g. Indentation, vertical spacing between methods, placement of braces, variable names, etc…) A maximum of five marks can be deducted if coding conventions is not followed.

INSTRUCTIONS ON HOW TO WRITE THE TEST ON ELECTRONIC CAMPUS (EC):

1. Create a folder on **D:\ Drive** and rename the folder your **student number**.
2. Save all your work in the folder you have created
3. Open a text editor (notepad++ is recommended) or MS Word and save your document in your student number folder.
4. Open the **browser**.
5. Type http://ec.tut.ac.za in the address field of the browser and press enter.
6. Click **Login.**
7. Type in student number for user name and if you are using EC for the first time you must use your student number as your password. (Not applicable any more. Please change to the current situation).
8. Alternatively, you must enter your password. (Same here).
9. Click **Login button**.
10. Click [**Continue**]
11. Select [**Student**]
12. Select [**Tests**]
13. Select [**Webtest 4**]
14. Type or Select **PPB115D/PPG115D/PPBF15D/PPGF15D** into the first text field.
15. Type or Select **20232** into the second text field.
16. Type or Select J into the third text field (The "**J**" must be a capital letter).
17. Click "**Next**"
18. Click on the "**Write**" button.
19. Click on "**[Start]**" and you will be in the test to submit your answers.
20. Click on **[Section]** on the test to submit each Questions, and click [Go] to submit your answer- Copy your answer and paste it in the space provided.
21. Click **[Test]** to move to the next or previous Questions.
22. Under each question click edit to update your answer.

Question 1 (True OR False) **[5 Marks]**

Preparation:

Log into EC and select from the list, your answers in the provided spaces for each question under **Section 1**.

	STATEMENT
1.1	A class header has an optional access modifier, the keyword class and an identifier/ class name.
1.2	In Java applications, all data members must always have a private access modifier.
1.3	Non-static methods are often referred to as class methods.
1.4	An accessor method can be used to change the instance variable of an object.
1.5	A class called Farm, has two data members **farmersName** and farmersAddress. The class has a mutator method that will receive the farmer's name. The code snippet below is the correct snippet to set the farmers name using the name sent to it. **public setFarmersName (String name)** **{** **farmersName = name;** **}**

Question 2 **[28]**

Concrete Classes

Instructions

Log on to EC and proceed to the correct test. Select **Section 2** and provide your answers to this question therein.

Preparations

- Create a folder in the **D** drive called **D:\My Documents\StudentNumber\Question2**, where **StudentNumber** is your actual student number.
- You will be supplied with two **text** files called **CarRentalApp** and **TestCarRentalApp**.
- Open the files using **Notepad++** and complete the question as instructed below.

Car Rental Application

XYZ is small business in Soshanguve which specializes in renting out vehicles to people. Currently, they have the following group of vehicles Hatch, Sedan, SUV and Vans. They want you to assist them with an application that will help with their rental services hiring services.

To successfully complete the application, follow the UML diagram shown in **Table 2.1,** the description of the application in **Table 2.2,** the partially completed code of **CarRentalApp** class under **Table 2.3**, and the partially completed code of **Test CarRentalApp** class under **Table 2.4**.

Table 2.1 UML diagram of CarRentalApp

CarRentalApp
- carGroup:String - ratePerDay: double - rentalDays: int
+ CarRentalApp(String, double, int) +getCarGroup(): String +getRatePerDay (): double +getRentalDays(): int + calculateRentalCost():double

Table 2.2 Description of CarRentalApp class

	Member/Method	**Description**	**Marks**
2.2.1	**carGroup** **ratePerDay** **rentalDays**	Instance fields.	1

2.2.2	**CarRentalApp()**	This method is a parameterised constructor, which must assign values to the instance variables using the values sent to it.	4
2.2.3	**getCarGroup()** **getRatePerDay()** **getRentalDays()**	Accessors methods	6
2.2.4	**calculateRentalCost()**	This method must calculate and return the rental cost using the duration of the rental and the rental rate for the car. This value must be inclusive of the 15% vat rate.	3

Create a concrete class called **CarRentalApp** that will accept the **vehicle group type**, **rental rate**, **duration** of the rental in days and calculate the rental cost including 15% vat rate. The program must then display the **vehicle group type**, **rental rate**, **duration** and the rental cost as indicated below in **Figure 2.1.**

The application's description:
Create an application called **TestCarRentalApp** that will work as per the given description in **Table 2.3** below.

Table 2.3: TestCarRentalApp description

	Method	**Description**	**Marks**
2.3.1	main()	Perform the following tasks inside the main method: 1. Prompt the user to enter the **vehicle group type**, **rental rate** and the **duratio**n for rental (**See Figure 2.1**). 2. Create an object called **rental** of the type **CarRentalApp** class using the parameterised constructor. 3. Display the **vehicle group type**, **rental rate**, the **duratio**n for rental and the rental cost (**See Figure 2.1**) by calling/invoking the correct methods.	14

Sample Output:

```
Enter the group type (Hatch,Sedan,SUV,Van): Van
Enter the rate per day: 255.54
Enter duration for the rental: 3
Car Model: Van
Rate Per Day: R255.54
Number of Days Rented: 3
Total Rental Cost: R881.61
```

Figure 2.1

Table 2.4: Partially completed code of the concrete class (CarRentalApp.java)

```
public class CarRentalApp {
    L       String carGroup;
    L       double ratePerDay;
    L       int rentalDays;

    // Methods here

    public  M       (String carGroups, double dayRate, int days) {
            N
            O
            P
}
    public  Q       calculateRentalCost() {
            R
    }
        public  S       getCarGroup()
        {
                T
        }
        public  U       getRatePerDay()
        {
                V
        }
        public  W       getRentalDays()
        {
                X
        }

}
```

Table 2.5: Partially completed code of the application class (TestCarRentalApp.java)

```
import java.util.Scanner;
A

public class TestCarRental{
   public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        B  df = new  B  ("00.00");

      // Prompt for rental information
      System.out.print("Enter the group type (Hatch, Sedan, SUV, Van): ");

      String rentalGroup =  C  ;

      System.out.print("Enter the rate per day: ");

      double rentalRate =  D  ;

      System.out.print("Enter duration for the rental: ");
      int  duration =  E  ;

      // Creating a rental object

        F  rental = new  G  ;

      // Displaying rental details and total cost

      System.out.println("Car Model: " +  H  );

      System.out.println("Rate Per Day: R" +  I  );

      System.out.println("Number of Days Rented: " +  J  );

      System.out.println("Total Rental Cost: R" +  K  );

   }
}
```